

# David A. Nichols, Ph.D.

## Address:

Department of Physics  
University of Virginia  
382 McCormick Road  
Charlottesville, VA, 22904, USA

**Telephone:** +1 (434) 924-0501

**Email:** [david.nichols@virginia.edu](mailto:david.nichols@virginia.edu)

**Webpage:** <https://dnichols1.github.io/>

**ORCID:** [0000-0002-4758-9460](https://orcid.org/0000-0002-4758-9460)

## Education

- Ph.D., Physics** *California Institute of Technology (Caltech)* 06/2012  
– Ph.D. thesis supervisor: Prof. Yanbei Chen
- M.S., Physics** *California Institute of Technology (Caltech)* 06/2008
- B.A., Mathematics and Physics** *Claremont McKenna College* 05/2006  
– Valedictorian, *summa cum laude*, honors in both degrees

## Academic Positions

- Associate Professor of Physics** *University of Virginia* 09/2025–present
- Assistant Professor of Physics** *University of Virginia* 08/2019–08/2025
- Senior Postdoctoral Researcher** *University of Amsterdam* 07/2018–07/2019  
– Postdoctoral supervisor: Prof. Samaya Nissanke
- Postdoctoral Researcher** *Radboud University* 09/2016–06/2018  
– Postdoctoral supervisor: Prof. Samaya Nissanke
- Postdoctoral Associate** *Cornell University* 09/2012–08/2016  
– Postdoctoral supervisor: Prof. Éanna É. Flanagan
- Postdoctoral Researcher** *Caltech* 06/2012–08/2012  
– Postdoctoral supervisor: Prof. Yanbei Chen

## Research Funding

- National Science Foundation, CAREER Award**, 12/2024–11/2029  
“Investigating Matter and Spacetime Using Gravitational Waves.” \$400,000
- National Science Foundation, Gravitation Theory**, 06/2023–05/2026  
“Dynamical and Strong-Field Gravitational-Wave Phenomenology.” \$210,000
- ORAU, Ralph E. Powe Faculty Enhancement Award**, 06/2021–05/2022  
“Searching for the gravitational-wave memory effect in LIGO and Virgo data.” \$10,000
- National Science Foundation, Gravitation Theory**, 09/2020–08/2023  
“Dynamical and Strong-Gravity Effects from Black-Hole Binaries.” \$210,000

## Courses Taught and Guest Lectures

### Courses Taught

- Intro to General Relativity (PHYS 5240)** *University of Virginia* Fall 2025

|                                                   |                                  |                           |
|---------------------------------------------------|----------------------------------|---------------------------|
| <b>Electricity &amp; Magnetism II (PHYS 7420)</b> | <i>University of Virginia</i>    | Spring 2023–2026          |
| <b>Theoretical Mechanics I (PHYS 7010)</b>        | <i>University of Virginia</i>    | Fall 2024                 |
| <b>Classical Mechanics (PHYS 3210)</b>            | <i>University of Virginia</i>    | Spring 2021–22, Fall 2023 |
| <b>Advanced General Relativity (PHYS 8240)</b>    | <i>University of Virginia</i>    | Fall 2019–2021            |
| <b>Computational Physics</b> (co-taught)          | <i>Cornell University</i>        | Spring 2015               |
| <b>Quantum Mechanics</b>                          | <i>Claremont McKenna College</i> | Spring 2011               |

## Guest Lectures

|                              |                                     |         |
|------------------------------|-------------------------------------|---------|
| <b>Astroparticle Physics</b> | <i>Amsterdam University College</i> | 05/2019 |
| <b>General Relativity</b>    | <i>University of Amsterdam</i>      | 04/2017 |

## Other Awards, Honors, and Fellowships

- David and Barbara Groce Graduate Fellowship, 2009–2010
- Rose Hills Foundation Graduate Fellowship, 2006–2007
- Barry M. Goldwater Scholarship, 2004–2006
- GGR Travel Grant, 2009, 2010, 2012
- Phi Beta Kappa, 2005

## Research Interests

Compact binaries, gravitational waves, gravitational-wave memory, asymptotic spacetime symmetries, electromagnetic counterparts of gravitational-wave sources, gravitational-wave tests of general relativity, black-hole perturbation theory, post-Newtonian theory, numerical relativity, black-hole physics.

## Professional Activities

### Advising

#### *Postdoctoral Scholars*

|                                                     |                               |                  |
|-----------------------------------------------------|-------------------------------|------------------|
| Nicholas Speeney (co-advised; ICI fellow)           | <i>University of Virginia</i> | starting 09/2026 |
| Indranil Chakraborty (Fulbright-Nehru fellow)       | <i>University of Virginia</i> | 10/2025–present  |
| Sayantani Datta (co-advised; Rising Scholar fellow) | <i>University of Virginia</i> | 08/2023–08/2025  |
| Alexander M. Grant                                  | <i>University of Virginia</i> | 09/2020–08/2023  |

#### *Ph.D. Students*

|                   |                               |                 |
|-------------------|-------------------------------|-----------------|
| Siddhant Siddhant | <i>University of Virginia</i> | 02/2022–present |
| Cuishan Liu       | <i>University of Virginia</i> | 05/2023–present |
| William Messman   | <i>University of Virginia</i> | 05/2026–present |
| Benjamin Wade     | <i>University of Virginia</i> | 02/2021–05/2026 |
| Arwa Elhashash    | <i>University of Virginia</i> | 07/2019–05/2025 |

#### *M.A./M.S. Students*

|                               |                           |                 |
|-------------------------------|---------------------------|-----------------|
| Oliver Boersma (co-advised)   | <i>Radboud University</i> | 01/2018–01/2020 |
| Andris Dorozsmai (co-advised) | <i>Radboud University</i> | 01/2017–07/2018 |

*Postbaccalaureate Students*

|                            |                               |                 |
|----------------------------|-------------------------------|-----------------|
| Benjamin Werchowsky ('25)  | <i>University of Virginia</i> | 09/2025–present |
| Leenie Wilcox (co-advised) | <i>University of Virginia</i> | 09/2023–08/2024 |

*Undergraduate Students (Research Projects)*

|               |                               |                 |
|---------------|-------------------------------|-----------------|
| Zachary Raney | <i>University of Virginia</i> | 01/2025–05/2025 |
| Bejoy Sen     | <i>University of Virginia</i> | 01/2022–05/2022 |
| Andrew Tuma   | <i>University of Virginia</i> | 09/2019–05/2020 |
| Yara Yousef   | <i>University of Virginia</i> | 09/2019–05/2020 |

**Scientific Collaboration Memberships**

|                                                                           |                 |
|---------------------------------------------------------------------------|-----------------|
| <b>LISA Consortium</b> , <i>Community member</i> ,                        | 05/2025–present |
| <b>Cosmic Explorer Consortium</b> , <i>Member</i> ,                       | 12/2020–present |
| <b>Athena Mission</b> , <i>Multimessenger Science Working Group</i>       | 02/2021–05/2025 |
| <b>LISA Consortium</b> , <i>Consortium member</i> ,                       | 01/2019–04/2025 |
| <b>Virgo Collaboration</b> , <i>Radboud and Nikhef groups</i> ,           | 09/2016–09/2019 |
| <b>GROWTH Collaboration</b> , <i>Radboud University group</i> ,           | 09/2017–09/2018 |
| <b>LIGO Scientific Collaboration</b> , <i>Caltech Relativity Theory</i> , | 09/2015–09/2016 |

**Ph.D. Thesis Committees**

|                                |                                                                      |         |
|--------------------------------|----------------------------------------------------------------------|---------|
| Victor Santos Guedes           | <i>University of Virginia (Physics)</i>                              | 06/2026 |
| Siddarth Ajith                 | <i>University of Virginia (Physics)</i>                              | 04/2026 |
| Nur E Mohammad Rifat           | <i>University of Virginia (Physics)</i>                              | 04/2026 |
| Deven Bhakta                   | <i>University of Virginia (Astronomy)</i>                            | 04/2026 |
| Alejandro de las Peñas Castaño | <i>University of Virginia (Mathematics)</i>                          | 04/2026 |
| Benjamin Wade                  | <i>University of Virginia (Physics)</i> , Chair/Supervisor           | 04/2026 |
| Kevin Cunningham               | <i>University College Dublin (Mathematics)</i> , External Examiner   | 10/2025 |
| Jessica Pascadlo               | <i>University of Virginia (Physics)</i>                              | 07/2025 |
| Arwa Elhashash                 | <i>University of Virginia (Physics)</i> , Chair/Supervisor           | 05/2025 |
| Kyle Corcoran                  | <i>University of Virginia (Astronomy)</i>                            | 08/2024 |
| Xinlun Cheng                   | <i>University of Virginia (Astronomy)</i>                            | 04/2024 |
| Shu-Yan (Vincent) Lau          | <i>University of Virginia (Physics)</i>                              | 03/2024 |
| Xiaoshan Huang                 | <i>University of Virginia (Astronomy)</i>                            | 05/2023 |
| Nan Jiang                      | <i>University of Virginia (Physics)</i>                              | 04/2023 |
| Ka Ho (Andy) Lam               | <i>University of Virginia (Astronomy)</i>                            | 07/2022 |
| Moritz Hübner                  | <i>Monash University (Physics and Astronomy)</i> , External Examiner | 09/2021 |
| Shammi Tahura                  | <i>University of Virginia (Physics)</i>                              | 07/2021 |
| Hannah Lewis                   | <i>University of Virginia (Astronomy)</i>                            | 06/2021 |
| Pallavi Patil                  | <i>University of Virginia (Astronomy)</i>                            | 12/2020 |
| Paul Zivick                    | <i>University of Virginia (Astronomy)</i>                            | 04/2020 |
| Christian Hayes                | <i>University of Virginia (Astronomy)</i>                            | 04/2020 |
| Zachary Carson                 | <i>University of Virginia (Physics)</i>                              | 03/2020 |

## Annual Ph.D. Review Committees

|                   |                         |              |
|-------------------|-------------------------|--------------|
| Josh Greaves      | <i>Committee Chair</i>  | 2026–present |
| Hayden Hollenbeck | <i>Committee Chair</i>  | 2025–present |
| Victor Guedes     | <i>Committee Member</i> | 2024–2026    |
| Nur Rifat         | <i>Committee Member</i> | 2024–2026    |
| Siddarth Ajith    | <i>Committee Member</i> | 2023–2026    |
| Jessica Pascadlo  | <i>Committee Chair</i>  | 2023–2025    |
| Vincent Lau       | <i>Committee Member</i> | 2021–2023    |
| Saikat Bera       | <i>Committee Chair</i>  | 2020–2023    |
| Nan Jiang         | <i>Committee Member</i> | 2020–2023    |
| Shammi Tahura     | <i>Committee Member</i> | 2020–2021    |

## Journal Guest Editing

|                                                                                      |                 |
|--------------------------------------------------------------------------------------|-----------------|
| <b>Classical and Quantum Gravity</b> , Guest Editor                                  | 07/2023–12/2025 |
| <i>Focus Issue: “Gravitational-Wave Memory Effects: From Theory to Observation.”</i> |                 |

## Conference and Seminar Organization

|                                                                                  |                 |
|----------------------------------------------------------------------------------|-----------------|
| <b>UVA Physics Colloquium and Hoxton Lecture</b> , Committee member,             | 08/2024–present |
| <b>UVA Physics Colloquium and Hoxton Lecture</b> , Committee chair,              | 08/2022–08/2024 |
| <b>UVA Gravity Seminar Series</b> , Committee chair,                             | 01/2021–08/2024 |
| <b>Gravitational Memory Effects (QMUL)</b> , Organizing committee,               | 06/2023         |
| <b>Holography and Gravitational Waves (IFPU)</b> , Organizing committee,         | 07/2022         |
| <b>GRAPPA Colloquium Series</b> , Organizing committee,                          | 09/2018–07/2019 |
| <b>UvA GW seminars</b> , Organizing committee,                                   | 09/2018–07/2019 |
| <b>72<sup>nd</sup> Netherlands Astronomy Conference</b> , Organizing committees, | 05/2017         |

## Peer Reviewing

**Journals:** Physical Review Letters, Physical Review D, SciPost Physics, Astrophysical Journal Letters, Astrophysical Journal, Journal of High Energy Physics, Classical and Quantum Gravity, General Relativity and Gravitation, Nature Communications, Proceedings of the Royal Society A, Journal of Cosmology and Astroparticle Physics, International Journal of Modern Physics D, European Physical Journal C

**Funding Agencies:** National Aeronautics and Space Administration (NASA), National Science Foundation (NSF), European Research Council (ERC), Science and Technologies Facilities Council (STFC), Comisión Nacional de Investigación Científico y Tecnológico (CONICYT)

## Publications

### Short-Author-List, Peer-Reviewed Articles

41. N. E. M. Rifat, **D. A. Nichols**, and K. Yagi, “Hybrid model for inspiral-merger-ringdown gravitational waveforms from comparable-mass, nonspinning binary black holes,” [Phys. Rev. D](#) **113**, 104051 (2026), [arXiv:2511.09646 \[gr-qc\]](#)

40. B. A. Wade and **D. A. Nichols**, “Intermediate mass-ratio inspirals in a dense dark-matter environment: Effects of the initial dark-matter distribution,” *Phys. Rev. D* **113**, 044014 (2026), [arXiv:2508.21132 \[gr-qc\]](#)
39. S. Tahura, **D. A. Nichols**, and K. Yagi, “Gravitational-wave memory effects in the Damour-Esposito-Farèse extension of Brans-Dicke theory,” *Phys. Rev. D* **112**, 084037 (2025), [arXiv:2501.07488 \[gr-qc\]](#)
38. A. Elhashash and **D. A. Nichols**, “Waveform models for the gravitational-wave memory effect: II. Time-domain and frequency-domain models for nonspinning binaries,” *Phys. Rev. D* **112**, 064014 (2025), [arXiv:2504.18635 \[gr-qc\]](#)
37. A. Elhashash and **D. A. Nichols**, “Waveform models for the gravitational-wave memory effect: Extreme mass-ratio limit and final memory offset,” *Phys. Rev. D* **111**, 044052 (2025), [arXiv:2407.19017 \[gr-qc\]](#)
36. E. Wilcox, **D. A. Nichols**, and K. Yagi, “Probing dark-matter effects with gravitational waves using the parameterized post-Einsteinian framework,” *Phys. Rev. D* **110**, 124009 (2024), [arXiv:2409.10846 \[gr-qc\]](#)
35. S. Siddhant, A. M. Grant, and **D. A. Nichols**, “Higher memory effects and the post-Newtonian calculation of their gravitational-wave signals,” *Classical Quantum Gravity* **41**, 205014 (2024), [arXiv:2403.13907 \[gr-qc\]](#)
34. E. E. Flanagan and **D. A. Nichols**, “Fully nonlinear transformations of the Weyl-Bondi-Metzner-Sachs asymptotic symmetry group,” *J. High Energy Phys.* **03**, 120 (2024), [arXiv:2311.03130 \[gr-qc\]](#)
33. **D. A. Nichols**, B. A. Wade, and A. M. Grant, “Secondary accretion of dark matter in intermediate mass-ratio inspirals: Dark-matter dynamics and gravitational-wave phase,” *Phys. Rev. D* **108**, 124062 (2023), [arXiv:2309.06498 \[gr-qc\]](#)
32. A. M. Grant and **D. A. Nichols**, “Outlook for detecting the gravitational wave displacement and spin memory effects with current and future gravitational wave detectors,” *Phys. Rev. D* **107**, 064056 (2023), [arXiv:2210.16266 \[gr-qc\]](#)
31. L. Piro, M. Ahlers, A. Coleiro, M. Colpi, E. de Oña Wilhelmi, M. Gianazzi, P. G. Jonker, P. McNamara, **D. A. Nichols**, *et al.*, “Multi-messenger-Athena Synergy White Paper,” *Exp. Astron.* **54**, 23 (2022), [arXiv:2110.15677 \[astro-ph.HE\]](#)
30. A. M. Grant and **D. A. Nichols**, “Persistent gravitational wave observables: Curve deviation in asymptotically flat spacetimes,” *Phys. Rev. D* **105**, 024056 (2022), [arXiv:2109.03832 \[gr-qc\]](#)
29. A. Coogan, G. Bertone, D. Gaggero, B. J. Kavanagh, and **D. A. Nichols**, “Measuring the dark matter environments of black hole binaries with gravitational waves,” *Phys. Rev. D* **105**, 043009 (2022), [arXiv:2108.04154 \[gr-qc\]](#)
28. S. Tahura, **D. A. Nichols**, and K. Yagi, “Gravitational-wave memory effects in Brans-Dicke theory: Waveforms and effects in the post-Newtonian approximation,” *Phys. Rev. D* **104**, 104010 (2021), [arXiv:2107.02208 \[gr-qc\]](#)
27. A. Elhashash and **D. A. Nichols**, “Definitions of (super) angular momentum in asymptotically flat spacetimes: Properties and applications to compact-binary mergers,” *Phys. Rev. D* **104**, 024020 (2021), [arXiv:2101.12228 \[gr-qc\]](#)
26. S. Tahura, **D. A. Nichols**, A. Saffer, L. C. Stein, and K. Yagi, “Brans-Dicke theory in Bondi-Sachs form: Asymptotically flat solutions, asymptotic symmetries and gravitational-wave memory effects,” *Phys. Rev. D* **103**, 104026 (2021), [arXiv:2007.13799 \[gr-qc\]](#)
25. B. J. Kavanagh, **D. A. Nichols**, G. Bertone, and D. Gaggero, “Detecting dark matter around black holes with gravitational waves: Effects of dark-matter dynamics on the gravitational waveform,” *Phys. Rev. D* **102**, 083006 (2020), [arXiv:2002.12811 \[gr-qc\]](#)
24. O. M. Boersma, **D. A. Nichols**, and P. Schmidt, “Forecasts for detecting the gravitational-

- wave memory effect with Advanced LIGO and Virgo,” *Phys. Rev. D* **101**, 083026 (2020), [arXiv:2002.01821 \[astro-ph.HE\]](#)
23. É. É. Flanagan, A. M. Grant, A. I. Harte, and **D. A. Nichols**, “Persistent gravitational wave observables: Nonlinear plane wave spacetimes,” *Phys. Rev. D* **101**, 104033 (2020), [arXiv:1912.13449 \[gr-qc\]](#)
  22. É. É. Flanagan, A. M. Grant, A. I. Harte, and **D. A. Nichols**, “Persistent gravitational wave observables: general framework,” *Phys. Rev. D* **99**, 084044 (2019), [arXiv:1901.00021 \[gr-qc\]](#)
  21. T. Hinderer, S. Nissanke, F. Foucart, K. Hotokezaka, T. Vincent, M. Kasliwal, P. Schmidt, A. R. Williamson, **D. A. Nichols**, M. Duez, L. E. Kidder, H. P. Pfeiffer, and M. A. Scheel, “Distinguishing the nature of comparable-mass neutron star binary systems with multimessenger observations: GW170817 case study,” *Phys. Rev. D* **100**, 063021 (2019), [arXiv:1808.03836 \[astro-ph.HE\]](#)
  20. **D. A. Nichols**, “Center-of-mass angular momentum and memory effect in asymptotically flat spacetimes,” *Phys. Rev. D* **98**, 064032 (2018), [arXiv:1807.08767 \[gr-qc\]](#)
  19. **D. A. Nichols**, “Spin memory effect for compact binaries in the post-Newtonian approximation,” *Phys. Rev. D* **95**, 084048 (2017), [arXiv:1702.03300 \[gr-qc\]](#)
  18. A. Ghosh, A. Ghosh, N. Johnson-McDaniel, C. K. Mitra, P. Ajith, W. Del Pozzo, **D. A. Nichols**, Y. Chen, A. B. Nielsen, C. P. L. Berry, and L. London, “Testing general relativity using golden black-hole binaries,” *Phys. Rev. D* **94**, 021101 (2016), [arXiv:1602.02453 \[gr-qc\]](#)
  17. É. É. Flanagan, **D. A. Nichols**, L. C. Stein, and J. Vines, “Prescriptions for measuring and transporting local angular momenta in general relativity,” *Phys. Rev. D* **93**, 104007 (2016), [arXiv:1602.01847 \[gr-qc\]](#)
  16. É. É. Flanagan and **D. A. Nichols**, “Conserved charges of the extended Bondi-Metzner-Sachs algebra,” *Phys. Rev. D* **95**, 044002 (2017), [arXiv:1510.03386 \[hep-th\]](#)
  15. J. Vines and **D. A. Nichols**, “Properties of an affine transport equation and its holonomy,” *Gen. Rel. Grav.* **48**, 127 (2016), [arXiv:1412.4077 \[gr-qc\]](#)
  14. É. É. Flanagan and **D. A. Nichols**, “Observer dependence of angular momentum in general relativity and its relationship to the gravitational-wave memory effect,” *Phys. Rev. D* **92**, 084057 (2015), [arXiv:1411.4599 \[gr-qc\]](#)
  13. R. H. Price, J. W. Belcher, and **D. A. Nichols**, “Comparison of electromagnetic and gravitational radiation: What we can learn about each from the other,” *Am. J. Phys.* **81**, 575 (2013), [arXiv:1212.4730 \[gr-qc\]](#)
  12. H. Yang, F. Zhang, A. Zimmerman, **D. A. Nichols**, E. Berti, and Y. Chen, “Branching of quasinormal modes for nearly extremal Kerr black holes,” *Phys. Rev. D* **87**, 041502 (2013), [arXiv:1212.3271 \[gr-qc\]](#)
  11. **D. A. Nichols**, A. Zimmerman, Y. Chen, G. Lovelace, K. D. Matthews, R. Owen, F. Zhang, and K. S. Thorne, “Visualizing Spacetime Curvature via Frame-Drag Vortexes and Tidal Tendexes III. Quasinormal Pulsations of Schwarzschild and Kerr Black Holes,” *Phys. Rev. D* **86**, 104028 (2012), [arXiv:1208.3038 \[gr-qc\]](#)
  10. F. Zhang, A. Zimmerman, **D. A. Nichols**, Y. Chen, G. Lovelace, K. D. Matthews, R. Owen, and K. S. Thorne, “Visualizing Spacetime Curvature via Frame-Drag Vortexes and Tidal Tendexes II. Stationary Black Holes,” *Phys. Rev. D* **86**, 084049 (2012), [arXiv:1208.3034 \[gr-qc\]](#)
  9. H. Yang, **D. A. Nichols**, F. Zhang, A. Zimmerman, Z. Zhang, and Y. Chen, “Quasinormal-mode spectrum of Kerr black holes and its geometric interpretation,” *Phys. Rev. D* **86**, 104006 (2012), [arXiv:1207.4253 \[gr-qc\]](#)
  8. **D. A. Nichols** and Y. Chen, “Hybrid method for understanding black-hole mergers: Inspiral case,” *Phys. Rev. D* **85**, 044035 (2012), [arXiv:1109.0081 \[gr-qc\]](#)

7. **D. A. Nichols.**, R. Owen, F. Zhang, A. Zimmermann, J. Brink, Y. Chen, G. Lovelace, K. D. Matthews, M. A. Scheel, and K. S. Thorne, “Visualizing Spacetime Curvature via Frame-Drag Vortexes and Tidal Tendexes I. General Theory and Weak-Gravity Applications,” *Phys. Rev. D* **84**, 124014 (2011), [arXiv:1108.5486 \[gr-qc\]](#)
6. A. Zimmerman, **D. A. Nichols**, and F. Zhang, “Classifying the Isolated Zeros of Asymptotic Gravitational Radiation by Tendex and Vortex Lines,” *Phys. Rev. D* **84**, 044037 (2011), [arXiv:1107.2959 \[gr-qc\]](#)
5. R. Owen, J. Brink, Y. Chen, J. D. Kaplan, G. Lovelace, K. D. Matthews, **D. A. Nichols**, M. A. Scheel, F. Zhang, A. Zimmerman, and K. S. Thorne, “Frame-Dragging Vortexes and Tidal Tendexes Attached to Colliding Black Holes: Visualizing the Curvature of Spacetime,” *Phys. Rev. Lett.* **106**, 151101 (2011), [arXiv:1012.4869 \[gr-qc\]](#)
4. **D. A. Nichols** and Y. Chen, “A hybrid method for understanding black-hole mergers: head-on case,” *Phys. Rev. D* **82**, 104020 (2010), [arXiv:1007.2024 \[gr-qc\]](#)
3. G. Lovelace, Y. Chen, M. Cohen, J. D. Kaplan, D. Keppel, K. D. Matthews, **D. A. Nichols**, M. A. Scheel, and U. Sperhake, “Momentum flow in black-hole binaries. II. Numerical simulations of equal-mass, head-on mergers with antiparallel spins,” *Phys. Rev. D* **82**, 064031 (2010), [arXiv:0907.0869 \[gr-qc\]](#)
2. D. Keppel, **D. A. Nichols**, Y. Chen, and K. S. Thorne, “Momentum Flow in Black Hole Binaries. I. Post-Newtonian Analysis of the Inspiral and Spin-Induced Bobbing,” *Phys. Rev. D* **80**, 124015 (2009), [arXiv:0902.4077 \[gr-qc\]](#)
1. J. D. Kaplan, **D. A. Nichols**, and K. S. Thorne, “Post-Newtonian Approximation in Maxwell-Like Form,” *Phys. Rev. D* **80**, 124014 (2009), [arXiv:0808.2510 \[gr-qc\]](#)

## Papers as Part of Scientific Collaborations

73. N. Afshordi *et al.* (LISA Consortium Waveform Working Group), “Waveform Modelling for the Laser Interferometer Space Antenna,” *Living Rev. Relativ.* **28**, 9 (2025), [arXiv:2311.01300 \[gr-qc\]](#)
72. K. G. Arun *et al.* (LISA), “New horizons for fundamental physics with LISA,” *Living Rev. Relativ.* **25**, 4 (2022), [arXiv:2205.01597 \[gr-qc\]](#)
71. B. P. Abbott *et al.* (KAGRA, LIGO Scientific, Virgo), “Prospects for observing and localizing gravitational-wave transients with Advanced LIGO, Advanced Virgo and KAGRA,” *Living Rev. Relativ.* **23**, 3 (2020), [arXiv:1304.0670 \[gr-qc\]](#)
70. R. Abbott *et al.* (LIGO Scientific, Virgo), “Search for Gravitational Waves Associated with Gamma-Ray Bursts Detected by Fermi and Swift During the LIGO-Virgo Run O3a,” *Astrophys. J.* **915**, 86 (2021), [arXiv:2010.14550 \[astro-ph.HE\]](#)
69. R. Abbott *et al.* (LIGO Scientific, Virgo), “Properties and Astrophysical Implications of the 150  $M_{\odot}$  Binary Black Hole Merger GW190521,” *Astrophys. J.* **900**, L13 (2020), [arXiv:2009.01190 \[astro-ph.HE\]](#)
68. R. Abbott *et al.* (LIGO Scientific, Virgo), “GW190521: A Binary Black Hole Merger with a Total Mass of 150  $M_{\odot}$ ,” *Phys. Rev. Lett.* **125**, 101102 (2020), [arXiv:2009.01075 \[gr-qc\]](#)
67. R. Abbott *et al.* (LIGO Scientific, Virgo), “GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object,” *Astrophys. J. Lett.* **896**, L44 (2020), [arXiv:2006.12611 \[astro-ph.HE\]](#)
66. R. Abbott *et al.* (LIGO Scientific, Virgo), “GW190412: Observation of a Binary-Black-Hole Coalescence with Asymmetric Masses,” *Phys. Rev. D* **102**, 043015 (2020), [arXiv:2004.08342 \[astro-ph.HE\]](#)
65. J. Broderick *et al.*, “LOFAR 144-MHz follow-up observations of GW170817,” *Mon. Not. Roy.*

- Astron. Soc.* **494**, 5110 (2020), [arXiv:2004.01726 \[astro-ph.HE\]](#)
64. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW190425: Observation of a Compact Binary Coalescence with Total Mass  $\sim 3.4M_{\odot}$ ,” *Astrophys. J. Lett.* **892**, L3 (2020), [arXiv:2001.01761 \[astro-ph.HE\]](#)
  63. R. Hamburg *et al.* (Fermi Gamma-ray Burst Monitor Team, LIGO Scientific, Virgo), “A Joint Fermi-GBM and LIGO/Virgo Analysis of Compact Binary Mergers From the First and Second Gravitational-wave Observing Runs,” *Astrophys. J.* **893**, 100 (2020), [arXiv:2001.00923 \[astro-ph.HE\]](#)
  62. F. Acernese *et al.* (Virgo), “The advanced Virgo longitudinal control system for the O2 observing run,” *Astropart. Phys.* **116**, 102386 (2020)
  61. F. Acernese *et al.* (Virgo), “Increasing the Astrophysical Reach of the Advanced Virgo Detector via the Application of Squeezed Vacuum States of Light,” *Phys. Rev. Lett.* **123**, 231108 (2019)
  60. R. Abbott *et al.* (LIGO Scientific, Virgo), “Open data from the first and second observing runs of Advanced LIGO and Advanced Virgo,” *SoftwareX* **13**, 100658 (2021), [arXiv:1912.11716 \[gr-qc\]](#)
  59. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “A guide to LIGO-Virgo detector noise and extraction of transient gravitational-wave signals,” *Classical Quantum Gravity* **37**, 055002 (2020), [arXiv:1908.11170 \[gr-qc\]](#)
  58. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo,” *Astrophys. J.* **909**, 218 (2021), [arXiv:1908.06060 \[astro-ph.CO\]](#)
  57. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “An Optically Targeted Search for Gravitational Waves emitted by Core-Collapse Supernovae during the First and Second Observing Runs of Advanced LIGO and Advanced Virgo,” *Phys. Rev. D* **101**, 084002 (2020), [arXiv:1908.03584 \[astro-ph.HE\]](#)
  56. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Model comparison from LIGO-Virgo data on GW170817’s binary components and consequences for the merger remnant,” *Classical Quantum Gravity* **37**, 045006 (2020), [arXiv:1908.01012 \[gr-qc\]](#)
  55. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for Eccentric Binary Black Hole Mergers with Advanced LIGO and Advanced Virgo during their First and Second Observing Runs,” *Astrophys. J.* **883**, 149 (2019), [arXiv:1907.09384 \[astro-ph.HE\]](#)
  54. B. P. Abbott *et al.* (LIGO Scientific, Virgo, IPN), “Search for gravitational-wave signals associated with gamma-ray bursts during the second observing run of Advanced LIGO and Advanced Virgo,” *Astrophys. J.* **886**, 75 (2019), [arXiv:1907.01443 \[astro-ph.HE\]](#)
  53. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for gravitational waves from Scorpius X-1 in the second Advanced LIGO observing run with an improved hidden Markov model,” *Phys. Rev. D* **100**, 122002 (2019), [arXiv:1906.12040 \[gr-qc\]](#)
  52. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for intermediate mass black hole binaries in the first and second observing runs of the Advanced LIGO and Virgo network,” *Phys. Rev. D* **100**, 064064 (2019), [arXiv:1906.08000 \[gr-qc\]](#)
  51. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “All-sky search for short gravitational-wave bursts in the second Advanced LIGO and Advanced Virgo run,” *Phys. Rev. D* **100**, 024017 (2019), [arXiv:1905.03457 \[gr-qc\]](#)
  50. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for sub-solar mass ultracompact binaries in Advanced LIGO’s second observing run,” *Phys. Rev. Lett.* **123**, 161102 (2019), [arXiv:1904.08976 \[astro-ph.CO\]](#)
  49. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “All-sky search for long-duration gravitational-

- wave transients in the second Advanced LIGO observing run,” *Phys. Rev. D* **99**, 104033 (2019), [arXiv:1903.12015 \[gr-qc\]](#)
48. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Directional limits on persistent gravitational waves using data from Advanced LIGO’s first two observing runs,” *Phys. Rev. D* **100**, 062001 (2019), [arXiv:1903.08844 \[gr-qc\]](#)
  47. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Tests of General Relativity with the Binary Black Hole Signals from the LIGO-Virgo Catalog GWTC-1,” *Phys. Rev. D* **100**, 104036 (2019), [arXiv:1903.04467 \[gr-qc\]](#)
  46. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “A search for the isotropic stochastic background using data from Advanced LIGO’s second observing run,” *Phys. Rev. D* **100**, 061101 (2019), [arXiv:1903.02886 \[gr-qc\]](#)
  45. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “All-sky search for continuous gravitational waves from isolated neutron stars using Advanced LIGO O2 data,” *Phys. Rev. D* **100**, 024004 (2019), [arXiv:1903.01901 \[astro-ph.HE\]](#)
  44. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Searches for Gravitational Waves from Known Pulsars at Two Harmonics in 2015-2017 LIGO Data,” *Astrophys. J.* **879**, 10 (2019), [arXiv:1902.08507 \[astro-ph.HE\]](#)
  43. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Narrow-band search for gravitational waves from known pulsars using the second LIGO observing run,” *Phys. Rev. D* **99**, 122002 (2019), [arXiv:1902.08442 \[gr-qc\]](#)
  42. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for Transient Gravitational-wave Signals Associated with Magnetar Bursts during Advanced LIGO’s Second Observing Run,” *Astrophys. J.* **874**, 163 (2019), [arXiv:1902.01557 \[astro-ph.HE\]](#)
  41. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Low-latency Gravitational-wave Alerts for Multimessenger Astronomy during the Second Advanced LIGO and Virgo Observing Run,” *Astrophys. J.* **875**, 161 (2019), [arXiv:1901.03310 \[astro-ph.HE\]](#)
  40. M. Soares-Santos *et al.* (DES, LIGO Scientific, Virgo), “First Measurement of the Hubble Constant from a Dark Standard Siren using the Dark Energy Survey Galaxies and the LIGO/Virgo Binary-Black-hole Merger GW170814,” *Astrophys. J.* **876**, L7 (2019), [arXiv:1901.01540 \[astro-ph.CO\]](#)
  39. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Searches for Continuous Gravitational Waves from 15 Supernova Remnants and Fomalhaut b with Advanced LIGO,” *Astrophys. J.* **875**, 122 (2019), [arXiv:1812.11656 \[astro-ph.HE\]](#)
  38. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs,” *Phys. Rev. X* **9**, 031040 (2019), [arXiv:1811.12907 \[astro-ph.HE\]](#)
  37. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Binary Black Hole Population Properties Inferred from the First and Second Observing Runs of Advanced LIGO and Advanced Virgo,” *Astrophys. J.* **882**, L24 (2019), [arXiv:1811.12940 \[astro-ph.HE\]](#)
  36. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Tests of General Relativity with GW170817,” *Phys. Rev. Lett.* **123**, 011102 (2019), [arXiv:1811.00364 \[gr-qc\]](#)
  35. A. Albert *et al.* (ANTARES, IceCube, LIGO Scientific, Virgo), “Search for Multi-messenger Sources of Gravitational Waves and High-energy Neutrinos with Advanced LIGO during its first Observing Run, ANTARES and IceCube,” *Astrophys. J.* **870**, 134 (2019), [arXiv:1810.10693 \[astro-ph.HE\]](#)
  34. E. Burns *et al.* (Fermi Gamma-ray Burst Monitor Team, LIGO Scientific, Virgo), “A Fermi Gamma-ray Burst Monitor Search for Electromagnetic Signals Coincident with Gravitational-Wave Candidates in Advanced LIGO’s First Observing Run,” *Astrophys. J.* **871**, 90 (2019),

- [arXiv:1810.02764 \[astro-ph.HE\]](#)
33. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for gravitational waves from a long-lived remnant of the binary neutron star merger GW170817,” *Astrophys. J.* **875**, 160 (2019), [arXiv:1810.02581 \[gr-qc\]](#)
  32. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Constraining the p-mode–g-mode tidal instability with GW170817,” *Phys. Rev. Lett.* **122**, 061104 (2019), [arXiv:1808.08676 \[astro-ph.HE\]](#)
  31. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for sub-solar mass ultracompact binaries in Advanced LIGO’s first observing run,” *Phys. Rev. Lett.* **121**, 231103 (2018), [arXiv:1808.04771 \[astro-ph.CO\]](#)
  30. M. Fishbach *et al.* (LIGO Scientific, Virgo), “A Standard Siren Measurement of the Hubble Constant from GW170817 without the Electromagnetic Counterpart,” *Astrophys. J. Lett.* **871**, L13 (2019), [arXiv:1807.05667 \[astro-ph.CO\]](#)
  29. F. Acernese *et al.* (Virgo), “Calibration of Advanced Virgo and Reconstruction of the Gravitational Wave Signal  $h(t)$  during the Science Run O2,” *Classical Quantum Gravity* **35**, 205004 (2018), [arXiv:1807.03275 \[gr-qc\]](#)
  28. L. Barack *et al.* (GWverse), “Black holes, gravitational waves and fundamental physics: a roadmap,” *Classical Quantum Gravity* **36**, 143001 (2019), [arXiv:1806.05195 \[gr-qc\]](#)
  27. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Properties of the binary neutron star merger GW170817,” *Phys. Rev. X* **9**, 011001 (2019), [arXiv:1805.11579 \[gr-qc\]](#)
  26. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170817: Measurements of neutron star radii and equation of state,” *Phys. Rev. Lett.* **121**, 161101 (2018), [arXiv:1805.11581 \[gr-qc\]](#)
  25. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “A Search for Tensor, Vector, and Scalar Polarizations in the Stochastic Gravitational-Wave Background,” *Phys. Rev. Lett.* **120**, 201102 (2018), [arXiv:1802.10194 \[gr-qc\]](#)
  24. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Full Band All-sky Search for Periodic Gravitational Waves in the O1 LIGO Data,” *Phys. Rev. D* **97**, 102003 (2018), [arXiv:1802.05241 \[gr-qc\]](#)
  23. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Constraints on cosmic strings using data from the first Advanced LIGO observing run,” *Phys. Rev. D* **97**, 102002 (2018), [arXiv:1712.01168 \[gr-qc\]](#)
  22. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170608: Observation of a 19-solar-mass Binary Black Hole Coalescence,” *Astrophys. J.* **851**, L35 (2017), [arXiv:1711.05578 \[astro-ph.HE\]](#)
  21. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for post-merger gravitational waves from the remnant of the binary neutron star merger GW170817,” *Astrophys. J.* **851**, L16 (2017), [arXiv:1710.09320 \[astro-ph.HE\]](#)
  20. M. M. Kasliwal *et al.* (GROWTH), “Illuminating Gravitational Waves: A Concordant Picture of Photons from a Neutron Star Merger,” *Science* **358**, 1559 (2017), [arXiv:1710.05436 \[astro-ph.HE\]](#)
  19. G. Hallinan *et al.* (GROWTH), “A Radio Counterpart to a Neutron Star Merger,” *Science* **358**, 1579 (2017), [arXiv:1710.05435 \[astro-ph.HE\]](#)
  18. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral,” *Phys. Rev. Lett.* **119**, 161101 (2017), [arXiv:1710.05832 \[gr-qc\]](#)
  17. B. P. Abbott *et al.* (LIGO Scientific, Virgo, GROND, SALT Group, OzGrav, DFN, INTEGRAL, Insight-Hxmt, MAXI Team, Fermi-LAT, J-GEM, RATIR, ATLAS, IceCube, CAAS-TRO, LWA, ePESSTO, GRAWITA, RIMAS, SKA South Africa/MeerKAT, H.E.S.S., 1M2H Team, IKI-GW Follow-up, Fermi GBM, Pi of Sky, DWF (Deeper Wider Faster Program),

- Dark Energy Survey, MASTER, AstroSat Cadmium Zinc Telluride Imager Team, Swift, Pierre Auger, ASKAP, VINROUGE, JAGWAR, Chandra Team at McGill University, TTU-NRAO, GROWTH, AGILE Team, MWA, ATCA, AST3, TOROS, Pan-STARRS, NuSTAR, BOOTES, CaltechNRAO, High Time Resolution Universe Survey, Nordic Optical Telescope, Las Cumbres Observatory Group, TZAC Consortium, LOFAR, IPN, DLT40, Texas Tech University, HAWC, ANTARES, KU, Dark Energy Camera GW-EM, CALET, Euro VLBI Team, ALMA), “Multi-messenger Observations of a Binary Neutron Star Merger,” *Astrophys. J.* **848**, L12 (2017), [arXiv:1710.05833 \[astro-ph.HE\]](#)
16. B. P. Abbott *et al.* (LIGO Scientific, Virgo, Fermi-GBM, INTEGRAL), “Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A,” *Astrophys. J.* **848**, L13 (2017), [arXiv:1710.05834 \[astro-ph.HE\]](#)
  15. B. P. Abbott *et al.* (LIGO Scientific, Virgo, VINROUGE, Las Cumbres Observatory, DLT40, 1M2H, MASTER, Dark Energy Camera GW-EM), “A gravitational-wave standard siren measurement of the Hubble constant,” *Nature* **551**, 85 (2017), [arXiv:1710.05835 \[astro-ph.CO\]](#)
  14. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Estimating the Contribution of Dynamical Ejecta in the Kilonova Associated with GW170817,” *Astrophys. J.* **850**, L39 (2017), [arXiv:1710.05836 \[astro-ph.HE\]](#)
  13. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170817: Implications for the Stochastic Gravitational-Wave Background from Compact Binary Coalescences,” *Phys. Rev. Lett.* **120**, 091101 (2018), [arXiv:1710.05837 \[gr-qc\]](#)
  12. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “On the Progenitor of Binary Neutron Star Merger GW170817,” *Astrophys. J.* **850**, L40 (2017), [arXiv:1710.05838 \[astro-ph.HE\]](#)
  11. A. Albert *et al.* (ANTARES, IceCube, Pierre Auger, LIGO Scientific, Virgo), “Search for High-energy Neutrinos from Binary Neutron Star Merger GW170817 with ANTARES, IceCube, and the Pierre Auger Observatory,” *Astrophys. J.* **850**, L35 (2017), [arXiv:1710.05839 \[astro-ph.HE\]](#)
  10. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “First narrow-band search for continuous gravitational waves from known pulsars in advanced detector data,” *Phys. Rev. D* **96**, 122006 (2017), [arXiv:1710.02327 \[gr-qc\]](#)
  9. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170814: A Three-Detector Observation of Gravitational Waves from a Binary Black Hole Coalescence,” *Phys. Rev. Lett.* **119**, 141101 (2017), [arXiv:1709.09660 \[gr-qc\]](#)
  8. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “First search for nontensorial gravitational waves from known pulsars,” *Phys. Rev. Lett.* **120**, 031104 (2018), [arXiv:1709.09203 \[gr-qc\]](#)
  7. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “First low-frequency Einstein@Home all-sky search for continuous gravitational waves in Advanced LIGO data,” *Phys. Rev. D* **96**, 122004 (2017), [arXiv:1707.02669 \[gr-qc\]](#)
  6. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “All-sky Search for Periodic Gravitational Waves in the O1 LIGO Data,” *Phys. Rev. D* **96**, 062002 (2017), [arXiv:1707.02667 \[gr-qc\]](#)
  5. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Upper Limits on Gravitational Waves from Scorpius X-1 from a Model-Based Cross-Correlation Search in Advanced LIGO Data,” *Astrophys. J.* **847**, 47 (2017), [arXiv:1706.03119 \[astro-ph.HE\]](#)
  4. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “GW170104: Observation of a 50-Solar-Mass Binary Black Hole Coalescence at Redshift 0.2,” *Phys. Rev. Lett.* **118**, 221101 (2017), [arXiv:1706.01812 \[gr-qc\]](#)
  3. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for intermediate mass black hole binaries in the first observing run of Advanced LIGO,” *Phys. Rev. D* **96**, 022001 (2017), [arXiv:1704.04628 \[gr-qc\]](#)

2. B. P. Abbott *et al.* (LIGO Scientific, Virgo), “Search for gravitational waves from Scorpius X-1 in the first Advanced LIGO observing run with a hidden Markov model,” *Phys. Rev. D* **95**, 122003 (2017), [arXiv:1704.03719 \[gr-qc\]](#)
1. F. Acernese *et al.* (Virgo), “Status of the Advanced Virgo gravitational wave detector,” *Int. J. Mod. Phys. A* **32**, 1744003 (2017)

### Conference Proceedings as Part of Scientific Collaborations

2. F. Acernese *et al.* (Virgo), “Status of Advanced Virgo,” *Proceedings, 6th International Conference on New Frontiers in Physics (ICNFP 2017): Crete, Greece, August 17-29, 2017*, *EPJ Web Conf.* **182**, 02003 (2018)
1. F. Acernese *et al.* (Virgo), “Status of the Advanced Virgo Gravitational Wave Detector,” in *Proceedings, International Conference on Cosmology, Gravitational Waves and Particles: Singapore, Singapore, February 6-10, 2017* (2018) pp. 1–12

### Short-Author-List, Non-Peer-Reviewed Articles

3. G. Compère and **D. A. Nichols**, “Classical and Quantized General-Relativistic Angular Momentum,” (2021), [arXiv:2103.17103 \[gr-qc\]](#)
2. A. Zimmerman, H. Yang, F. Zhang, **D. A. Nichols**, E. Berti, and Y. Chen, “Reply to ‘On the branching of quasinormal resonances of near-extremal Kerr black holes’ by Shahar Hod,” (2015), [arXiv:1510.08159 \[gr-qc\]](#)
1. **D. A. Nichols**, “Frontiers of Neutron Star Astrophysics,” in *Matters of Gravity, The Newsletter of the Topical Group on Gravitation of the American Physical Society*, Vol. 44, edited by D. Garfinkle (2014) [arXiv:1412.8368 \[gr-qc\]](#)

### Ph.D. Thesis

1. **D. A. Nichols**, *Visualizing, approximating, and understanding black-hole binaries*, [Ph.D. thesis](#), Caltech (2012)

|                                                |
|------------------------------------------------|
| <h2>Talks and Conference Sessions Chaired</h2> |
|------------------------------------------------|

### Invited Talks

40. **IFPU Workshop, Perspectives on Gravity**, 06/2026, Trieste, Italy  
“Gravitational-wave memory effects from binary-black-hole mergers.”
39. **U. South Dakota, Physics Seminar**, 03/2026, Vermillion, SD, USA  
“Gravitational-wave signatures of dark matter around black holes.”
38. **James Madison U., Astronomy & Physics Seminar**, 02/2026, Harrisonburg, VA, USA  
“Studying gravity at its extremes with gravitational waves.”
37. **UIUC PAX-X Workshop and CE Symposium**, 07/2025, Urbana-Champaign, IL, USA  
“Panel discussion on dark matter.” (Panelist)
36. **GGI Asymptotic Symmetries Workshop**, 06/2025, Florence, Italy  
“Detection prospects of gravitational-wave memory effects from binary black holes.”
35. **UPenn, HET Seminar**, 12/2024, Philadelphia, PA, USA  
“Gravitational-wave memory effects from binary-black-hole mergers.”
34. **IRCHEP 1403, Plenary Session 2**, 10/2024, Online  
“Gravitational-wave memory effects from astrophysical sources.”

33. **University of Virginia Physics Colloquium**, *09/2024*, Charlottesville, VA, USA  
“Gravitational waves: A window into matter and gravity at their extremes.”
32. **Simons Celestial Holography Workshop**, *04/2024*, New York, NY, USA  
“Gravitational-wave memory effects as an observational window into celestial holography.”
31. **U. Helsinki, Cosmology Seminar**, *09/2023*, Online  
“Gravitational-wave memory effects from binary-black-hole mergers.”
30. **Niels Bohr Institute, LISA FP Workshop**, *08/2023*, Copenhagen, Denmark  
“Dark matter and primordial black holes: Perspectives from the working group.”
29. **University College Dublin, ACM Seminar**, *06/2023*, Dublin, Ireland  
“Gravitational-wave memory effects from binary-black-hole mergers.”
28. **UMass Dartmouth, CSCDR Seminar**, *03/2023*, Dartmouth, MA, USA  
“Gravitational-wave memory effects from binary-black-hole mergers.”
27. **Holography and Gravitational Waves Workshop**, *07/2022*, Trieste, Italy  
“Prospects for observations of gravitational-wave memory and celestial holography.”
26. **33rd Rencontre de Blois**, *05/2022*, Blois, France  
“Gravitational-wave signatures of dark matter around black holes.”
25. **University of Virginia Physics Colloquium**, *01/2022*, Charlottesville, VA, USA  
“Studying matter and spacetime with gravitational waves.”
24. **Celestial Amplitudes Workshop, Discussion Panel**, *09/2021*, Online  
“Observing gravitational-wave memory effects and the infrared triangle.”
23. **Lyon/Marseille Summer School**, *06/2021*, Online  
“Gravitational waves: Testing classical (and quantum?) gravity.”
22. **Perimeter Institute Strong Gravity Seminar**, *04/2021*, Online  
“Gravitational-wave memory effects from binary-black-hole mergers.”
21. **Harvard Black-Hole Initiative Colloquium**, *12/2020*, Online  
“Gravitational-wave memory effects from binary-black-hole mergers.”
20. **Athena Multimessenger Synergy Workshop**, *05/2020*, Online  
“Multimessenger observations with future gravitational-wave observatories.”
19. **Virginia Tech Neutrino Physics Seminar**, *10/2019*, Blacksburg, VA, USA  
“Gravitational-wave memory effects from binary-black-hole mergers.”
18. **Birmingham University Astrophysics Seminar**, *05/2019*, Birmingham, UK  
“Computing and detecting gravitational-wave memory effects from binary black holes.”
17. **University of Virginia Physics Colloquium**, *02/2019*, Charlottesville, VA, USA  
“Gravitational waves and fundamental properties of matter and spacetime.”
16. **University of Mississippi Physics Colloquium**, *02/2019*, Oxford, MS, USA  
“Gravitational waves and fundamental properties of matter and spacetime.”
15. **Southern Methodist University Physics Seminar**, *02/2019*, Dallas, TX, USA  
“Gravitational waves and fundamental properties of matter and spacetime.”
14. **Athena Multimessenger Workshop**, *11/2018*, Alicante, Spain  
“Prospects for multimessenger observations with future gravitational-wave observatories.”
13. **MITP Sound of Spacetime Workshop**, *06/2018*, Mainz, Germany  
“Gravitational-wave memory effects: Observables and prospects for measurement.”
12. **ULB Solvay Workshop, Infrared Physics**, *05/2018*, Brussels, Belgium  
“Gravitational-wave memory effects: Observables and prospects for measurement.”
11. **Perimeter Institute, Strong-Gravity Seminar**, *02/2018*, Waterloo, Canada  
“Gravitational waves: Exploring the strongly curved side of the Universe.”
10. **University of Guelph, Physics Colloquium**, *02/2018*, Guelph, Canada

- “Gravitational waves: Exploring the strongly curved side of the Universe.”
9. **Jena University, Quantum & Gravitational Fields Seminar**, 01/2018, Jena, Germany  
“Spin memory effect and charges of the extended BMS algebra.”
  8. **Nikhef, PAX Workshop**, 08/2017, Amsterdam, Netherlands  
“Gravitational-wave memory effects: Testing general relativity by measuring non-oscillatory gravitational waves from binary black holes.”
  7. **Nordita, Physics of Extreme-Gravity Stars**, 06/2017, Stockholm, Sweden  
“Testing general relativity with the LIGO observations of binary black holes.”
  6. **AEI Postdam, Astrophysical Relativity Seminar**, 12/2016, Postdam, Germany  
“Gravitational-wave memory observables and charges of the extended BMS algebra”
  5. **University of Southampton Gravity Seminar**, 11/2016, Southampton, UK  
“Gravitational-wave memory observables and charges of the extended BMS algebra”
  4. **NPCSM Workshop, YITP, Kyoto University**, 11/2016, Kyoto, Japan  
“Gravitational-wave memory observables and charges of the extended BMS algebra”
  3. **Kavli IPMU, Mathematics–String-Theory Seminar**, 10/2016, Tokyo, Japan  
“Gravitational-wave memory observables and charges of the extended BMS algebra”
  2. **Radboud IMAPP Quantum Gravity Seminar**, 10/2016, Nijmegen, Netherlands  
“Gravitational-wave memory observables and charges of the extended BMS algebra”
  1. **Claremont Colleges Mathematics Colloquium**, 02/2011, Claremont, CA, USA  
“Black-Hole Binaries: Observing, Visualizing, and Understanding Strongly Curved, Dynamical Spacetime”

## Contributed Talks

22. **APS April Meeting**, 04/2023, Minneapolis, MN, USA  
“Gravitational-wave signatures of dark-matter capture in intermediate mass-ratio inspirals.”
21. **APS April Meeting**, 04/2022, New York City, NY, USA  
“Approximate inspiral-merger-ringdown waveforms for black-hole mergers in extensions of general relativity.”
20. **APS April Meeting**, 04/2021, Online  
“Effects of dynamical dark-matter distributions on intermediate mass-ratio inspirals (IMRIs).”
19. **LISA Symposium XIII**, 09/2020, Online  
“Effects of dynamical dark-matter distributions on IMRI and EMRI systems.”
18. **American Physical Society April Meeting**, 04/2020, Online  
“Forecasts for detecting the gravitational-wave memory effect with Advanced LIGO and Virgo.”
17. **GWverse Global Meeting**, 01/2018, Valetta, Malta  
“Gravitational-wave spin memory effect for compact binaries.”
16. **Physics of Extreme-Gravity Stars**, 06/2017, Stockholm, Sweden  
“Gravitational-wave displacement and spin memory effects.”
15. **6<sup>th</sup> Dutch-Belgian GW Meeting**, 03/2017, Leuven, Belgium  
“Spin memory effect for compact binaries”
14. **GrAMPa Workshop**, 08/2016, Paris, France  
“Gravitational-wave memory observables”
13. **GR21 Meeting**, 07/2016, New York City, NY, USA  
“Conserved charges of the extended Bondi-Metzner-Sachs algebra”
12. **American Physical Society April Meeting**, 04/2016, Salt Lake City, UT, USA  
“Methods for measuring and transporting angular momentum in general relativity”

11. **Eastern Gravity Meeting, 05/2015,** Rochester, NY, USA  
“Subleading gravitational-wave memory effects”
10. **Eastern Gravity Meeting, 05/2014,** Morgantown, WV, USA  
“Ambiguity in angular momentum and its relationship to gravitational-wave memory”
9. **American Physical Society April Meeting, 04/2014,** Savannah, GA, USA  
“Ambiguity in angular momentum and its relationship to gravitational-wave memory”
8. **American Physical Society April Meeting, 04/2012,** Atlanta, GA, USA  
“Tendex and Vortex Lines of Perturbed Schwarzschild and Kerr Black Holes”
7. **Pacific Coast Gravity Meeting, 03/2012,** Santa Barbara, CA, USA  
“Tendex and Vortex Lines of Black-Hole Spacetimes”
6. **American Physical Society April Meeting, 04/2011,** Anaheim, CA, USA  
“Vortex and Tendex Lines in Post-Newtonian and Black-Hole Perturbation Spacetimes”
5. **Pacific Coast Gravity Meeting, 03/2011,** Pasadena, CA, USA  
“Vortex and Tendex Lines in Post-Newtonian and Black-Hole Perturbation Spacetimes”
4. **Pacific Coast Gravity Meeting, 03/2010,** San Diego, CA, USA  
“A Hybrid Approximation Technique for Head-on Black-Hole-Binary Mergers”
3. **American Physical Society April Meeting, 02/2010,** Washington DC, USA  
“A Hybrid Approximation Technique for Head-on Binary-Black-Hole Mergers”
2. **American Physical Society April Meeting, 05/2009,** Denver, CO, USA  
“Momentum Flow in Inspiring Binary Black Holes”
1. **Pacific Coast Gravity Meeting, 03/2009,** Eugene, OR, USA  
“Momentum Flow in Black-Hole Binaries: Post-Newtonian Approximation”

## Sessions Chaired

5. **Gravitational Memory Effects (QMUL), 06/2023,** London, UK  
“Afternoon Session 1: Memory effects and observables”
4. **American Physical Society April Meeting, 04/2021,** Online  
“Session T17: Topics in Classical General Relativity”
3. **2017 Dutch Astronomy Conference, 05/2017,** Nijmegen, Netherlands  
“Plenary Session 6”
2. **Fall 2016 NOVA NW3 Meeting, 09/2016,** Nijmegen, Netherlands  
“Session 1: Compact Objects”
1. **American Physical Society April Meeting, 04/2016,** Salt Lake City, UT, USA  
“Session X18: Dynamics and Observables of Curved Spacetime”

## Outreach Talks

7. **Hoos in STEM Podcast, 06/2024,** Charlottesville, VA, USA  
“Personal Stories with Story Collider”
6. **Hoos in STEM Podcast, 11/2023,** Charlottesville, VA, USA  
“UVA’s Physicists Are Studying Black Holes and Unraveling Ancient Paradoxes”
5. **Radboud Huygens Colloquium, 10/2017,** Nijmegen, Netherlands  
“The discovery of a binary neutron star merger”
4. **ICMS Nobel Prizes 2017 Evening, 10/2017,** Eindhoven, Netherlands  
“The 2017 Nobel Prize in Physics”
3. **Cornell Astro REU Gravitational Waves Workshop, 06/2015,** Ithaca, NY, USA  
“Detecting Gravitational Waves with Laser Interferometry”

- 
2. **Cornell Astro REU Gravitational Waves Workshop**, *07/2014*, Ithaca, NY, USA  
“Detecting Gravitational Waves with Laser Interferometry”
  1. **Cornell Astro REU Gravitational Waves Workshop**, *07/2013*, Ithaca, NY, USA  
“Detecting Gravitational Waves with Laser Interferometry”